Coalition withholding: specialist review brief

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Review request. Please assess the correctness and priority of *The cost of coalition withholding: resource dimension and coalition size*. The central sparse-production claim is an unrestricted arity-three lower of order $s^{1/2}$. The dimension frontiers and sparse upper are consolidated in the same manuscript with full proofs. A graph section supplies a $1 + 1/\sqrt{2}$ full-OUT construction on complete P4 blow-ups and disjoint unions. We seek specific invalid steps, missing hypotheses, or antecedent theorems that imply the bounds. No claim of confirmed historical novelty or a major-open-problem resolution is being made.

1. Freeze this contract before comparing results

An institution fixes finite actual owners, resource coordinates, public technology F, and coordinatewise report boxes $0 \le b_i \le \bar{b}_i$. All real reports, including zero, are legal. Write $b_C = \sum_{i \in C} b_i$. Nonnegative payments satisfy, at every profile,

$$y_C(b) \ge F(b_C) \quad \text{for every coalition } C, \qquad \sum_i y_i(b) \le \kappa F(b_N).$$

For $|C| \le s$ and $0 \le \delta_i \le b_i$ supported on C, the contracts are

$$\text{TCWM: } y_C(b) - y_C(b - \delta) \ge 0; \qquad \text{full OUT: } y_C(b) - y_C(b - \delta) \ge F(\delta_C).$$

Unchanged coalition members are allowed. The scalar κ bounds *total funding*; the external supplement is $(\kappa - 1)F(b_N)$. There is no dummy-zero axiom, marginal floor, anonymity, continuity, linearity, or efficient-price representation imposed on a lower-bound mechanism. Owners are never cloned during a deviation.

The dimension class consists of finite-valued monotone concave homogeneous F on d resource types, with every coalition protected. The sparse class is finite packing production $F(c) = \max\{\pi \cdot x : Ax \le c,\ x \ge 0\}$, $A, \pi \ge 0$, with at most r positive resource entries per profitable column. Both n and d may grow; only incentive coalitions have size at most s.

2. Claims and where to audit them

Claim	Bound / qualification	Manuscript location
Universal dimension	$D_d = (d+1)/2$; $D_d^{\mathrm{OUT}} = 1 + c(d-1)$, $c(2 - e^{-1/c}) = 1$	Section 3
Sparse upper	$K_r(s) \le K_r^{\mathrm{OUT}}(s) \le rs^{r-1}$; full OUT, mixed owners	Section 4
Arity-three lower	$K_3(s) \ge \sqrt{s/6}$ for arbitrary rules; all real fixed boxes	Section 5
Sharper finite certificate	$K_3(s) \ge B_s \sim \sqrt{2\pi s}/(3\sqrt{e})$; $B_6 = 493/243$	Section 5.3
Protected-leaf graph rule	$1+1/\sqrt{2}$ full OUT on unit-profit P4, complete blow-ups and disjoint unions; fixed mixed owners	Section 6
Higher arity / pure owners	$\Omega_r(s^{(r-2)/(r-1)})$; separate pure realization	Appendix A

The arity-three gap to $3s^2$ remains open. The graph theorem is not a bound for arbitrary graphs and does not establish optimality even on P4. The optimized ladder length is optimal only within this certificate. The witnesses have exponentially many resource rows; no fixed-dimension or polynomial-size lower is claimed.

3. The sparse lower-bound proof in one page

Take nonzero vectors of $\mathbb{F}_2^{M+1}$ as resource rows, with basis $v, e_1, \dots, e_M$. Activities are all triples $\{a, b, a+b\}$, with unit inputs and profits. There are $2M+1$ fully eligible actual owners, each with real report box $[0,1]^{2^{M+1}-1}$. At profile C_m the cap owner reports v, every high owner reports e_j, and low owner j reports $v + e_j$ only when $j \le m$. Dormant high reports stay present. Exactly the active branch triples are supported, so production is one.

Finite averaging over legal resource and owner permutations preserves every contract inequality. The seed stabilizers exchange the cap with either private owner at C_1, giving $q_1 \ge 1/3$ by coverage. Active private owners are transitive at C_m, with payment q_m. Restoring low owner m while protecting it together with $k_m = \min\{s-1, 2m-2\}$ old active private owners gives

$$(k_m + 1)q_m \ge k_m q_{m-1}.$$

The old restorer's payment is discarded by nonnegativity, never assumed zero. The cap and a branch satisfy $z + 2q_M \ge 1$, while total payment is at least $z + 2Mq_M$. Hence

$$\kappa \ge 1 + \frac{2(M-1)}{3} \prod_{m=2}^{M} \frac{k_m}{k_m + 1}.$$

The uncapped product is governed by $Q_M = 4^M/(2\binom{2M}{M})$; it proves the uniform square-root lower. For fixed $s \ge 2$, the capped certificate is maximized at $M = s, s+1$. All coefficients can be cancelled with explicit nonnegative rational multipliers. For an asymmetric original rule the expensive profile can be an orbit image of C_M, rather than the displayed labeling.

The pure-owner version fixes one owner per resource, permits only that owner's scalar report, and averages the *joint* resource/owner automorphisms. Its count is $n = d = 2^{M+1} - 1$, not $2M+1$. Higher arity replaces triples by minimal binary circuits and retains the same logic.

Graph construction: a separate audit target. For the path B–A–D–C, form the directed bipartite double cover with left capacities unchanged and right capacities scaled by $\alpha = 1/\sqrt{2}$. Among maximum flows, clamp each leaf's left use near half its capacity within the feasible interval. Collapsing directed flow yields the funding bound $(1 + \alpha)F$. The difficult step is the selector: every coalition restoration has nonnegative derivative coefficients whose sums across each graph edge are at least one. A direct generic-perturbation segment proof treats all ties and zero coordinates. Exact transportation within complete groups transfers the result to blow-ups without changing actual owners. Full details are in manuscript Section 6; the general-tree extension is not a theorem of this draft. Section6 also proves an exact P6 obstruction to layerwise leaf clamping for every fixed reserve fraction below one and every scaling $0 < \alpha < 1$. This is a failure of that explicit recursion, not an impossibility for arbitrary tree rules.

4. Correctness questions with the highest consequence

1. For the graph selector, do the two maximum-flow intervals and all piecewise formulas agree, do the holding-cube inequalities cover every cell, and does the boundary argument preserve full OUT for passive and mixed owners? Does the complete-group transport preserve the withheld bundle's value as well as current production?
2. Is the fixed-institution averaging lawful on the entire real report box, including all dormant and zero-report owners? In the pure version, does the joint action preserve every individual eligibility restriction?
3. Do the seed and private-owner stabilizers preserve the whole profile, and are the stated activities the only supported columns of the full LP?
4. Does the coalition-size accounting allow every used transition without silently imposing a dummy-zero or individual-incumbent condition?
5. Is the finite-owner compactness argument for the OUT dimension lower valid without regularity assumptions on the original payments? In particular, check inactive payments, dilation averaging and shifted abundant totals.
6. Do the face-average uppers handle activation and discontinuities at support boundaries? For the sparse upper, check the coefficient signs for $|C| \leq s$ and coverage at the same-domain withheld profile.

5. Priority questions and established ingredients

Strong group monotonicity in Moulin–Sprumont's 2004 author version is a close aggregate-sum antecedent; its Proposition 2(ii) additionally imposes additivity, separability and exact balance. Doğan's 2013 absence-proofness working paper is a close binary-participation analogue of OUT for efficient subgames. Immorlica–Mahdian–Mirrokni's profit-sharing lower uses per-incumbent cross-monotonicity. Those distinctions prevent a direct reading of the inspected results as the present theorem, but do not establish priority.

Static cost of stability, including projective square-root constructions, bounded-rank positive hypergraphs, and binomial marginal values are established. Our claimed object is a quantitative cross-report funding obstruction despite an efficient static cover at each packing profile. Please identify any theorem that already yields it after a legitimate change of variables or normalization. In particular:

1. Do any subsidized absence-proofness or strong-group-monotonicity results already imply either dimension endpoint or the sparse coalition-size lower?
2. Is there a prior unrestricted lower with a better exponent for fixed arity, under this same aggregate incentive contract and all-coalition coverage?

3. Do the unpublished-to-published comparisons change the contract or conclusion? The packet leaves Sprumont's nearly-serial, serial-axiomatization and discrete Aumann–Shapley comparisons, and Doğan's published version, explicitly unresolved.
4. Are there applicable results in Schoppmann's small-coalition cost-sharing work or Moulin–Sprumont's cross-subsidization paper? These are additional leads, not claims of a theorem transfer.

The graph comparison also needs scrutiny: Xiao–Fang's inspected 2021 preprint uses efficient, per-incumbent population-monotonic matching allocations, whereas our subsidized rule protects aggregate coalition payments under divisible reports. Does any related result imply the quantitative selector bound under this exact contract? The flow/cut and transportation ingredients are classical; novelty is not assigned to them.

The detailed source matrix records exact versions and locators. Missing originals were not purchased; no unsuccessful access attempt is treated as clearance. Specialist suggestions for overlooked terminology and sources are particularly valuable.

6. Packet, checks and requested outcome

Read the current manuscript supplied with this dated packet and `review/PRIORITY_MATRIX.txt`. Use the file-version information in `README.txt`; older 18 October PDF exports omit the graph section. The source is standalone `manuscript.tex`. Run `python3 reproduce.py` in the package for the selected exact checks; `review/REPRODUCTION_NOTES.txt` gives the precise ranges. The six inherited checks use only the Python standard library. For the graph diagnostics, copy both graph checker files together into a scratch directory, then run `protected_leaf_exact_replay.py` and `leaf_recursion_exact_check.py` with Python. The first checks 625 exact algebraic profiles and holding-cube minima; the second checks the displayed recursion obstruction. Neither checks all real profiles, and both write adjacent JSON output. Assertions must remain enabled. Finite checks support the written proof; they do not establish the all-size theorem or historical novelty.

A useful review outcome is a list of fatal gaps, repairable steps, exact antecedents, and an assessment of whether the results merit one coherent paper. The initial mathematical readers should collectively cover cooperative production/core theory, cost-sharing incentives, and algorithmic game theory. The packet is prepared for human review; no person has been contacted or represented as having endorsed this draft.